EECP 臨床試驗設計分析報告

Analysis Report on EECP Clinical Trial Design

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日期 (Date): 2026年2月9日 (Feb 09, 2026)

摘要

本報告旨在詳細分析由林教授團隊所提出，關於「增強型體外反搏治療 (EECP) 對高血壓前期及第一期高血壓患者血壓之療效與劑量反應」的臨床試驗設計。此試驗採用一個創新的兩階段、隨機、雙盲、偽治療對照 (sham-controlled) 設計，旨在嚴謹地評估 EECP 療法的有效性及其最佳治療劑量。本報告將深入解析其試驗流程的各個環節，並從方法學角度探討其設計的優點與潛在的缺點及挑戰，最後提供綜合評估。

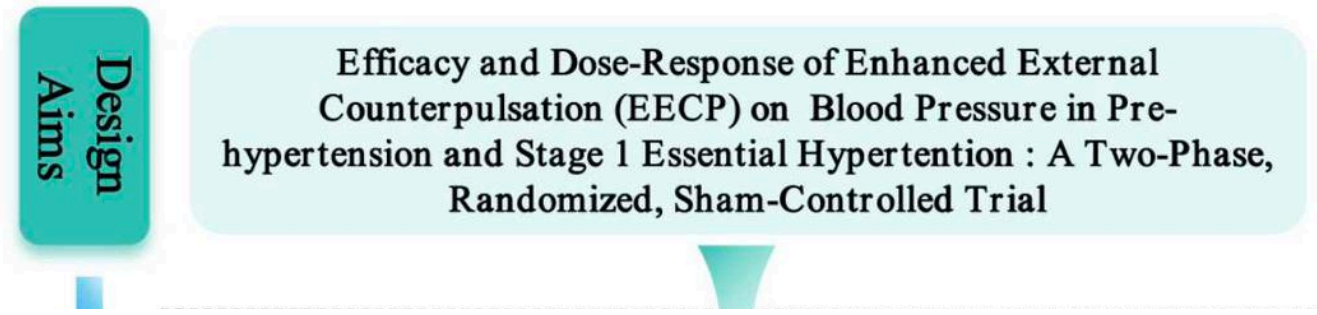
Abstract

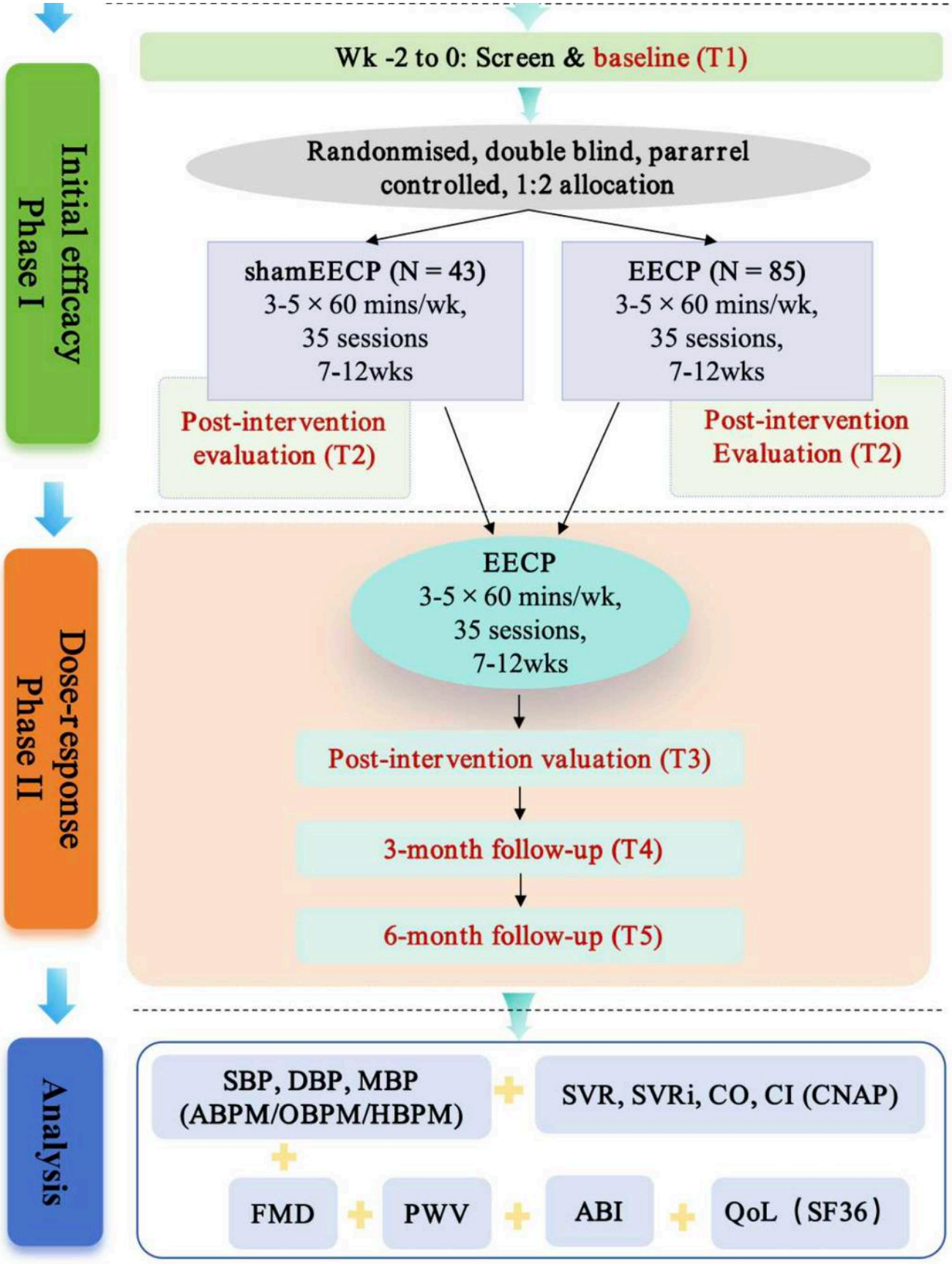
This report provides a detailed analysis of the clinical trial design proposed by Professor Lam's team concerning the "Efficacy and Dose-Response of Enhanced External Counterpulsation (EECP) on Blood Pressure in Pre-hypertension and Stage 1 Essential Hypertension." The trial employs an innovative two-phase, randomized, double-blind, sham-controlled design to rigorously evaluate the efficacy of EECP therapy and determine its optimal dosage. This report will deconstruct the trial protocol, examine the methodological advantages of its design, discuss potential disadvantages and challenges, and offer a comprehensive overall assessment.

臨床試驗流程圖 / Clinical Trial Flowchart

以下為林教授團隊提出的臨床試驗流程圖原圖：

The following is the original clinical trial flowchart proposed by Professor Lam's team:





圖說 (Figure Caption): 「增強型體外反搏 (EECP) 對高血壓前期及第一期原發性高血壓患者血壓之療效與劑量反應：一項兩階段、隨機、偽治療對照試驗」之研究設計流程圖。圖中清楚展

> 示了從篩選 (T1)、第一階段隨機對照 (T2)、第二階段劑量反應 (T3)，到三個月 (T4) 及六個月 (T5) 追蹤的完整試驗路徑，以及最終的分析指標。

> **Figure Caption:** Flowchart of the study design for "Efficacy and Dose-Response of Enhanced External Counterpulsation (EECP) on Blood Pressure in Pre-hypertension and Stage 1 Essential Hypertension: A Two-Phase, Randomized, Sham-Controlled Trial." The chart clearly illustrates the complete trial pathway from screening (T1), Phase I randomized controlled comparison (T2), Phase II dose-response (T3), to 3-month (T4) and 6-month (T5) follow-ups, along with the final analysis metrics.

臨床試驗設計解析 / Clinical Trial Design Analysis

根據上方的流程圖，該臨床試驗的設計核心在於其分階段且對照嚴謹的方法。以下將其結構拆解說明：

Based on the flowchart above, the core of this clinical trial design lies in its phased and rigorously controlled methodology. The structure is broken down as follows:

階段 (Phase)	項目 (Item)	描述 (Description)
設計目標 (Design Aims)	**研究主題 (Topic)**	評估 EECP 對高血壓前期及第一期原發性高血壓患者血壓的療效與劑量反應 (Efficacy and Dose-Response of EECP on Blood Pressure in Pre-hypertension and Stage 1 Essential Hypertension)。
篩選與基線 (Screen & Baseline)	**時間 (Timeline)**	第 -2 週至第 0 週 (Week -2 to 0)。
	內容 (Content)	篩選合適的受試者並建立基線數據 (T1)。
第一階段 (Phase I: Initial Efficacy)	**設計 (Design)**	隨機、雙盲、平行對照，以 1:2 的比例分配 (Randomised, double blind, parallel controlled, 1:2 allocation)。
	分組 (Groups)	1. **偽治療組 (shamEECP Group):** N=43 2. **EECP 治療組 (EECP Group):** N=85
	治療方案 (Intervention)	兩組均進行為期 7-12 週、共 35 次、每次 60 分鐘的療程，每週

		3-5 次。偽治療組接受極低壓力 (如 75 mmHg) 的反搏，以達到致盲效果 1 。 (Both groups undergo 35 sessions of 60 minutes each, 3-5 times per week for 7-12 weeks. The sham group receives very low pressure (e.g., 75 mmHg) to maintain blinding 1 .)
	評估 (Evaluation)	介入後評估 (T2) (Post-intervention evaluation (T2))。
第二階段 (Phase II: Dose-Response)	**設計 (Design)**	開放標籤 (Open-label)，所有第一階段的參與者均接受標準 EECP 治療。 (All participants from Phase I receive standard EECP treatment.)
	治療方案 (Intervention)	為期 7-12 週、共 35 次、每次 60 分鐘的 EECP 療程，每週 3-5 次。 (35 sessions of 60-minute EECP, 3-5 times per week for 7-12 weeks.)
	評估 (Evaluation)	1. 介入後評估 (T3) (Post-intervention evaluation (T3)) 2. 三個月後追蹤 (T4) (3-month follow-up (T4)) 3. 六個月後追蹤 (T5) (6-month follow-up (T5))
分析指標 (Analysis)	**血壓與血流動力學 (BP & Hemodynamics)**	SBP, DBP, MBP (透過 ABPM/OBPM/HBPM 測量), SVR, SVRi, CO, CI (透過 CNAP 測量)。
	血管功能與其他 (Vascular Function & Others)	FMD (內皮功能), PWV (動脈硬度), ABI (周邊動脈疾病), QoL (生活品質, SF36 問卷)。

設計優點分析 / Analysis of Advantages

此試驗設計展現了高度的科學嚴謹性，其優點主要體現在以下幾個方面：

This trial design demonstrates a high degree of scientific rigor, with its advantages primarily reflected in the following aspects:

1. 隨機雙盲與偽治療對照 (Randomized, Double-Blind, Sham-Controlled Design)

這是評估非藥物治療 (特別是涉及物理設備的治療) 的黃金標準。**隨機分派**能最大限度地減少選擇性偏差，確保兩組在基線時具有可比性。**雙盲設計** (研究者與受試者均不知曉分組情況) 結合**偽治療 (Sham Control)**，是此設計最強大的部分。偽治療組接受看似相同但無實際治療效果的程序 (例如，使用遠低於治療壓力的反搏)，這能有效控制安慰劑效應以及因接受治療這一行為本身所帶來的心理與生理影響。歷史上的 MUST-EECP 試驗就成功採用了此方法，以 75 mmHg 的低壓力作為偽治療，證實了其可行性與科學價值 [1]。這使得研究結果能更準確地歸因於 EECP 本身的生理效應，而非其他混雜因素。

This is the gold standard for evaluating non-pharmacological treatments, especially those involving physical devices. **Randomization** minimizes selection bias and ensures comparability between groups at baseline. The **double-blind design** combined with a **sham control** is the most powerful aspect of this protocol. The sham group undergoes a procedure that appears identical but has no actual therapeutic effect (e.g., using counterpulsation at a pressure far below the therapeutic threshold), effectively controlling for the placebo effect and the psychological and physiological impact of the treatment procedure itself. The landmark MUST-EECP trial successfully employed this method, using 75 mmHg low-pressure as the sham, confirming its feasibility and scientific value [1].

2. 兩階段設計 (Two-Phase Design)

這種設計非常巧妙。第一階段直接比較 EECP 與偽治療的療效，是回答「EECP 是否有效？」這一核心問題的關鍵。而第二階段讓所有參與者 (包括原偽治療組) 都接受標準 EECP 治療，這不僅解決了倫理考量 (讓所有參與者最終都能獲得潛在的有效治療)，更重要的是，它允許研究者觀察：(a) 原偽治療組在接受真正治療後的反應 (Dose-Response)；(b) 原治療組繼續接受治療或停止治療後的長期效果。這種設計能提供比單純平行對照試驗更豐富的數據，有助於理解療效的持續性與劑量關係。

This design is highly ingenious. Phase I directly compares EECP with sham treatment to answer the core question: "Is EECP effective?" Phase II then allows all participants (including the original sham group) to receive standard EECP treatment. This not only addresses ethical considerations (ensuring all participants eventually receive potentially effective treatment) but, more importantly, enables researchers to observe: (a) the response of the original sham group after receiving real treatment (Dose-Response); (b) the long-term effects on the original treatment group. This design yields richer data than a simple parallel-controlled trial.

3. 1:2 的不對等分配比例 (1:2 Allocation Ratio)

將較多的受試者 (85人) 分配到 EECP 治療組，而較少者 (43人) 分配到偽治療組，這種不對等分配有其特定考量。主要目的可能是為了收集更多關於實驗性治療 (EECP) 的安全性、耐受性及療效變

異性的數據。同時，從招募角度看，告知潛在受試者有更高機率 (2/3 的機會) 進入真正的治療組，可能會增加其參與意願 [2]。

Assigning more subjects (85) to the EECP group and fewer (43) to the sham group has specific considerations. The primary purpose is likely to collect more data on the safety, tolerability, and efficacy variability of the experimental treatment. From a recruitment perspective, informing potential subjects of a higher probability (2/3 chance) of entering the real treatment group may increase their willingness to participate [2].

4. 全面的分析指標 (Comprehensive Analysis Metrics)

此試驗不僅僅關注傳統的血壓讀數 (SBP, DBP)，還納入了多維度的評估指標。**動態血壓監測 (ABPM)** 能提供更全面的24小時血壓變化資訊。**血流動力學參數 (SVR, CO, CI)**、**血管功能指標 (FMD, PWV)** 以及 **生活品質 (QoL)** 的測量，使得研究能從多個層面深入探討 EECP 的作用機制，例如它是否透過改善血管內皮功能、降低動脈硬度或影響全身血管阻力來達到降壓效果。這極大地提升了研究的深度與科學價值。

The trial goes far beyond traditional blood pressure readings (SBP, DBP), incorporating multi-dimensional assessment metrics. **Ambulatory Blood Pressure Monitoring (ABPM)** provides comprehensive 24-hour blood pressure profiles. The measurement of **hemodynamic parameters (SVR, CO, CI)**, **vascular function indicators (FMD, PWV)**, and **Quality of Life (QoL)** enables a multi-faceted investigation into EECP's mechanisms of action, greatly enhancing the depth and scientific value of the study.

設計缺點與潛在挑戰 / Analysis of Disadvantages and Potential Challenges

儘管設計嚴謹，此試驗仍面臨一些固有的缺點與挑戰：

Despite its rigorous design, the trial faces several inherent disadvantages and challenges:

1. 偽治療的倫理與實施挑戰 (Ethical and Practical Challenges of Sham Control)

雖然偽治療在科學上是必要的，但它也引發倫理爭議，因為偽治療組的參與者需投入大量時間 (長達12週，共35小時) 卻無法獲得直接的治療益處。研究團隊必須在知情同意過程中極其詳盡地解釋這一點。此外，要完美實現雙盲並不容易。受試者可能會透過感受到的壓力差異猜測自己所在的分組，儘管 MUST-EECP 試驗證明低壓力可作為有效的偽治療，但完全消除感知差異仍是挑戰 [1]。

Although sham treatment is scientifically necessary, it raises ethical concerns because participants in the sham group must invest significant time (up to 12 weeks, 35 hours total) without receiving direct therapeutic benefit. Additionally, achieving perfect double-blinding is not easy. Subjects may guess their group assignment through perceived pressure differences, and while the MUST-EECP trial demonstrated low-pressure sham as feasible, completely eliminating perceptual differences remains a challenge [1].

2. 1:2 分配比例的統計學代價 (Statistical Cost of 1:2 Allocation)

雖然不對等分配有其優點，但也需付出代價。相較於 1:1 的分配，1:2 的分配比例會降低統計檢定力 (statistical power)，這意味著需要更大的總樣本量才能偵測到相同的治療效果。一篇分析文章指出，2:1 的分配比例會使所需的總樣本數增加約 12.5% [2]。如果 EECP 的真實療效不如預期中顯著，這種設計可能會增加出現偽陰性結果 (即未能偵測到真實存在的療效) 的風險。此外，如果實驗性治療最終被證明無效甚至有害，那麼 2:1 的分配會導致多 50% 的患者暴露於無效治療中 [2]。

While unequal allocation has its merits, it comes at a cost. Compared to 1:1 allocation, a 1:2 ratio reduces statistical power, meaning a larger total sample size is needed to detect the same treatment effect. An analysis article noted that a 2:1 allocation ratio increases the required total sample size by approximately 12.5% [2]. Furthermore, if the experimental treatment ultimately proves ineffective or even harmful, the 2:1 allocation results in 50% more patients being exposed to the ineffective treatment [2].

3. 試驗的複雜性與成本 (Complexity and Cost)

兩階段設計、長時間的追蹤以及多樣化的檢測指標，都使得此試驗在執行上非常複雜且成本高昂。需要精密的協調管理，以確保各階段數據收集的一致性與完整性，並維持受試者在長達六個月追蹤期內的高遵循度 (compliance)。受試者的脫落 (dropout) 可能會對研究結果的有效性構成威脅。

The two-phase design, extended follow-up, and diverse outcome measures make the trial highly complex and costly to execute. Meticulous coordination is required to ensure consistency and completeness of data collection across phases, and to maintain high participant compliance throughout the six-month follow-up period. Subject dropout could threaten the validity of the study results.

4. 結果的外部效度 (External Validity)

該試驗的受試者是高血壓前期及第一期高血壓患者，其結果是否能推廣至更嚴重的高血壓患者或其他合併症 (如糖尿病、冠心病) 的人群，尚需後續研究。嚴格的納入與排除標準雖然能提高內部效度，但可能會限制研究結果的普遍適用性。

The trial targets patients with pre-hypertension and Stage 1 hypertension. Whether the results can be generalized to patients with more severe hypertension or other comorbidities (e.g., diabetes, coronary heart disease) requires further investigation. While strict inclusion and exclusion criteria enhance internal validity, they may limit the generalizability of the findings.

優缺點總覽表 / Summary Table of Advantages and Disadvantages

類別 (Category)	優點 (Advantages)	缺點 (Disadvantages)
研究設計 (Study Design)	隨機雙盲偽治療對照為黃金標準，有效排除安慰劑效應與偏差 (Gold standard RCT with sham control eliminates placebo effect and bias)	偽治療組受試者投入大量時間卻無直接治療益處，引發倫理爭議 (Sham group participants invest significant time without direct benefit, raising ethical concerns)
兩階段結構 (Two-Phase Structure)	兼顧療效驗證與劑量反應探索，並解決倫理問題 (Addresses both efficacy verification and dose-response exploration while resolving ethical issues)	試驗執行複雜度高、成本高昂、脫落風險增加 (High operational complexity, cost, and increased dropout risk)
1:2 分配比例 (1:2 Allocation)	收集更多治療組安全性數據，可能提高招募意願 (More safety data on treatment arm; may improve recruitment)	統計檢定力降低，總樣本量需增加約 12.5%；若治療無效，更多患者暴露於風險中 (Reduced statistical power; 12.5% larger sample needed; more patients exposed if treatment is ineffective)
分析指標 (Outcome Measures)	涵蓋血壓、血流動力學、血管功能及生活品質，全面深入 (Comprehensive coverage of BP, hemodynamics, vascular function, and QoL)	多指標增加數據管理與分析的複雜性 (Multiple metrics increase data management and analytical complexity)
致盲 (Blinding)	偽治療使用低壓力反搏，經 MUST-EECP 驗證可行 (Low-pressure sham validated by MUST-EECP trial)	受試者可能透過壓力感受差異猜測分組 (Subjects may guess assignment through pressure perception differences)
外部效度 (External Validity)	嚴格標準提高內部效度，結果可靠性高 (Strict criteria enhance internal validity and result reliability)	結果未必能推廣至更嚴重的高血壓或合併症患者 (Results may not generalize to more severe hypertension or comorbid populations)

總結 / Conclusion

總體而言，林教授團隊提出的這項 EECP 臨床試驗設計是一個經過深思熟慮、科學嚴謹且具有創新性的研究方案。它巧妙地結合了偽治療對照、雙盲、兩階段設計以及不對等隨機分配，旨在以

極高的標準評估 EECP 的降壓療效及其作用機制。其全面的評估指標更為深入理解 EECP 的生理效應提供了可能。

然而，該設計也伴隨著倫理考量、統計效率的權衡以及執行上的高度複雜性。偽治療的實施與 1:2 分配比例的統計代價是其最主要的挑戰。若能成功克服這些挑戰，此項研究的成果將極具價值，不僅能為 EECP 在高血壓治療領域的應用提供堅實的循證醫學證據，也將為未來非藥物療法的臨床試驗設計提供一個重要的參考範例。

Overall, the EECP clinical trial design proposed by Professor Lam's team is a well-conceived, scientifically rigorous, and innovative research protocol. It skillfully combines a sham control, double-blinding, a two-phase structure, and unequal randomization to evaluate the antihypertensive efficacy of EECP and its mechanisms to a very high standard. The comprehensive set of outcome measures further enables a deep understanding of EECP's physiological effects.

However, the design is also accompanied by ethical considerations, trade-offs in statistical efficiency, and a high degree of operational complexity. The implementation of the sham control and the statistical cost of the 1:2 allocation ratio represent its most significant challenges. If these challenges can be successfully overcome, the results of this study will be of immense value. It will not only provide robust evidence-based support for the application of EECP in hypertension management but also serve as an important reference model for the design of future clinical trials for non-pharmacological therapies.

參考文獻 / References

[1] Arora, R. R., et al. (1999). The multicenter study of enhanced external counterpulsation (MUST-EECP): effect of EECP on exercise-induced myocardial ischemia and anginal episodes. Journal of the American College of Cardiology, 33(7), 1833-1840.

[2] Freidlin, B., & Korn, E. L. (2024). Two-to-One Randomization: Rarely Advisable. JCO Oncology Practice, 20(12), 1555-1558.